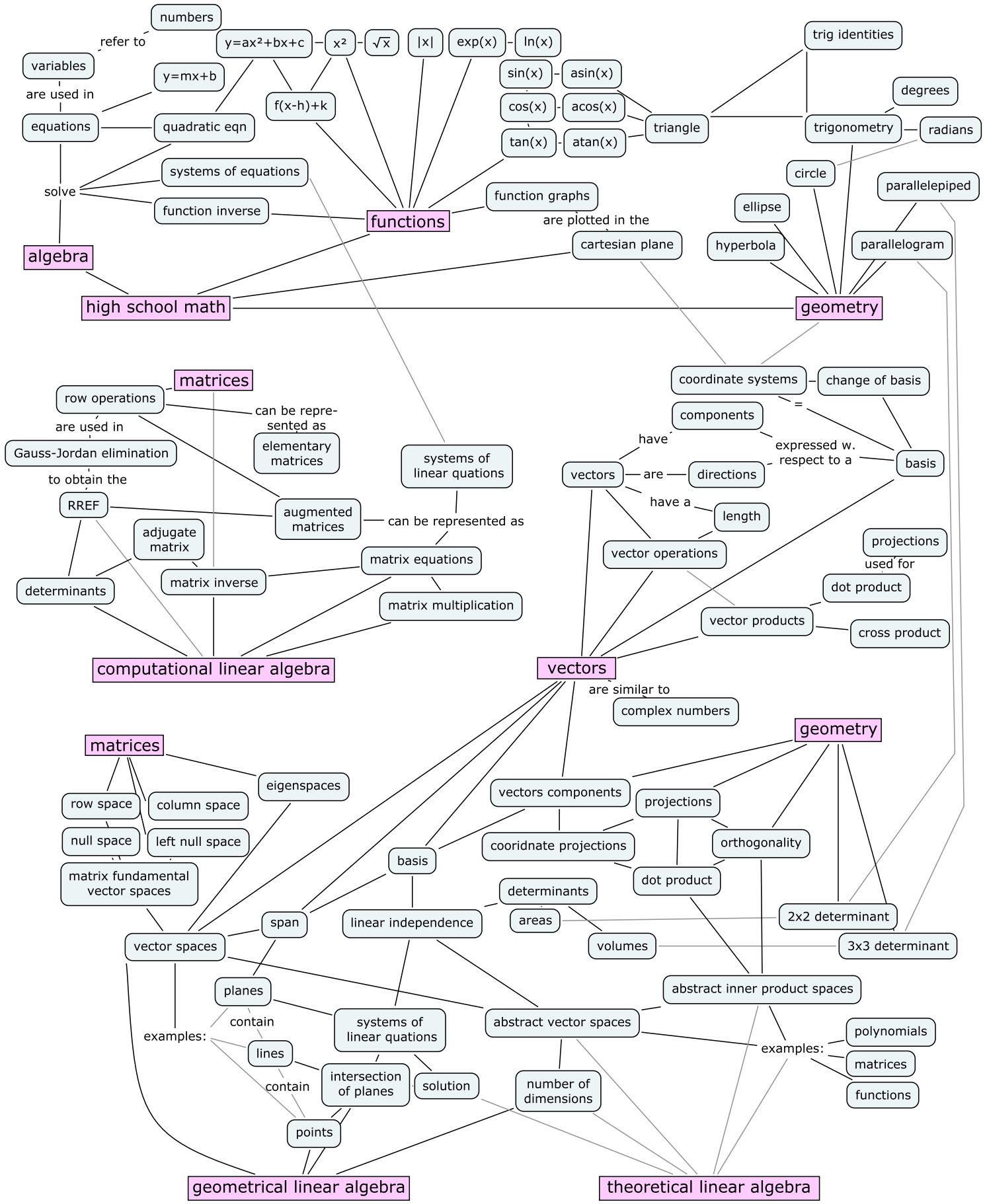
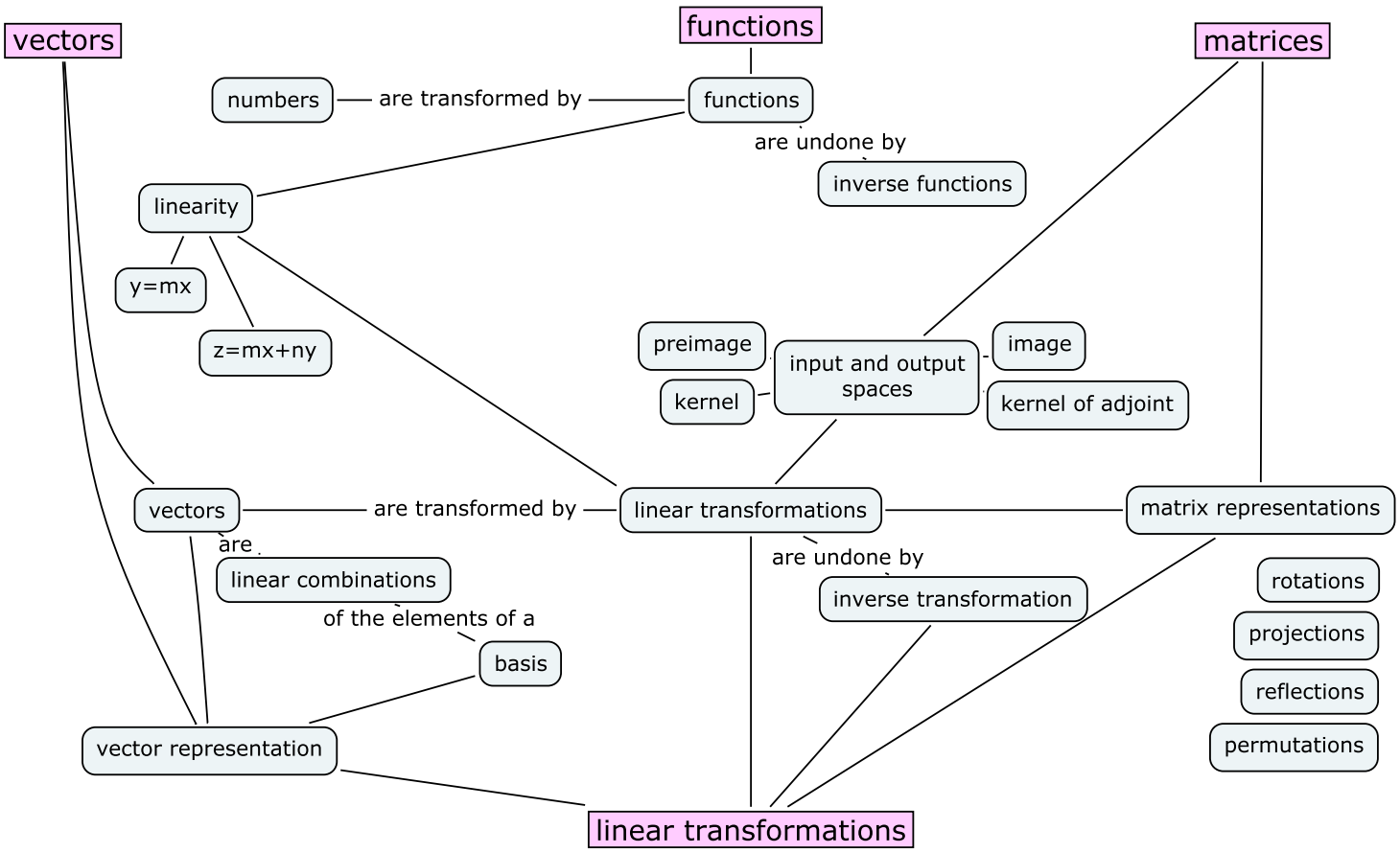
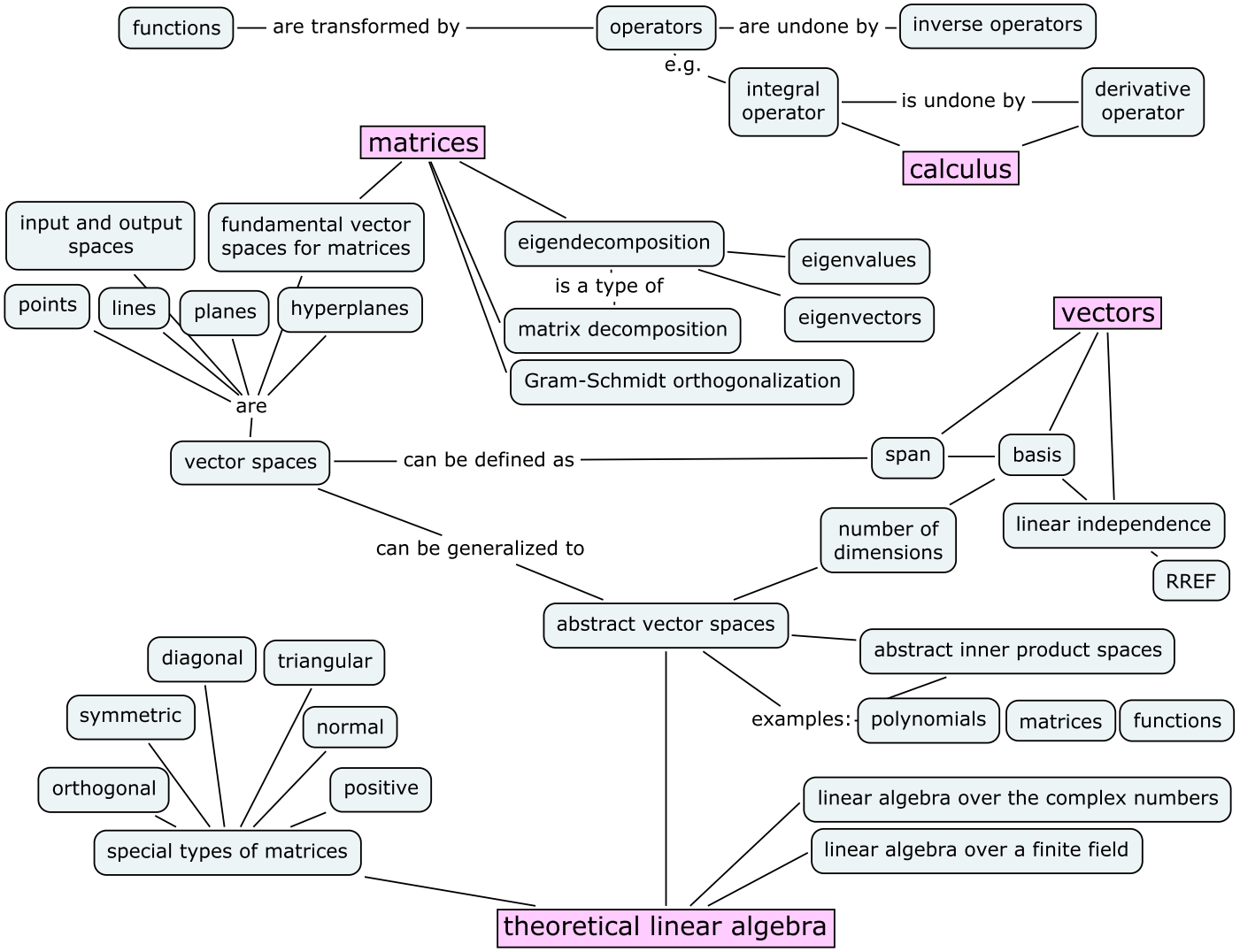
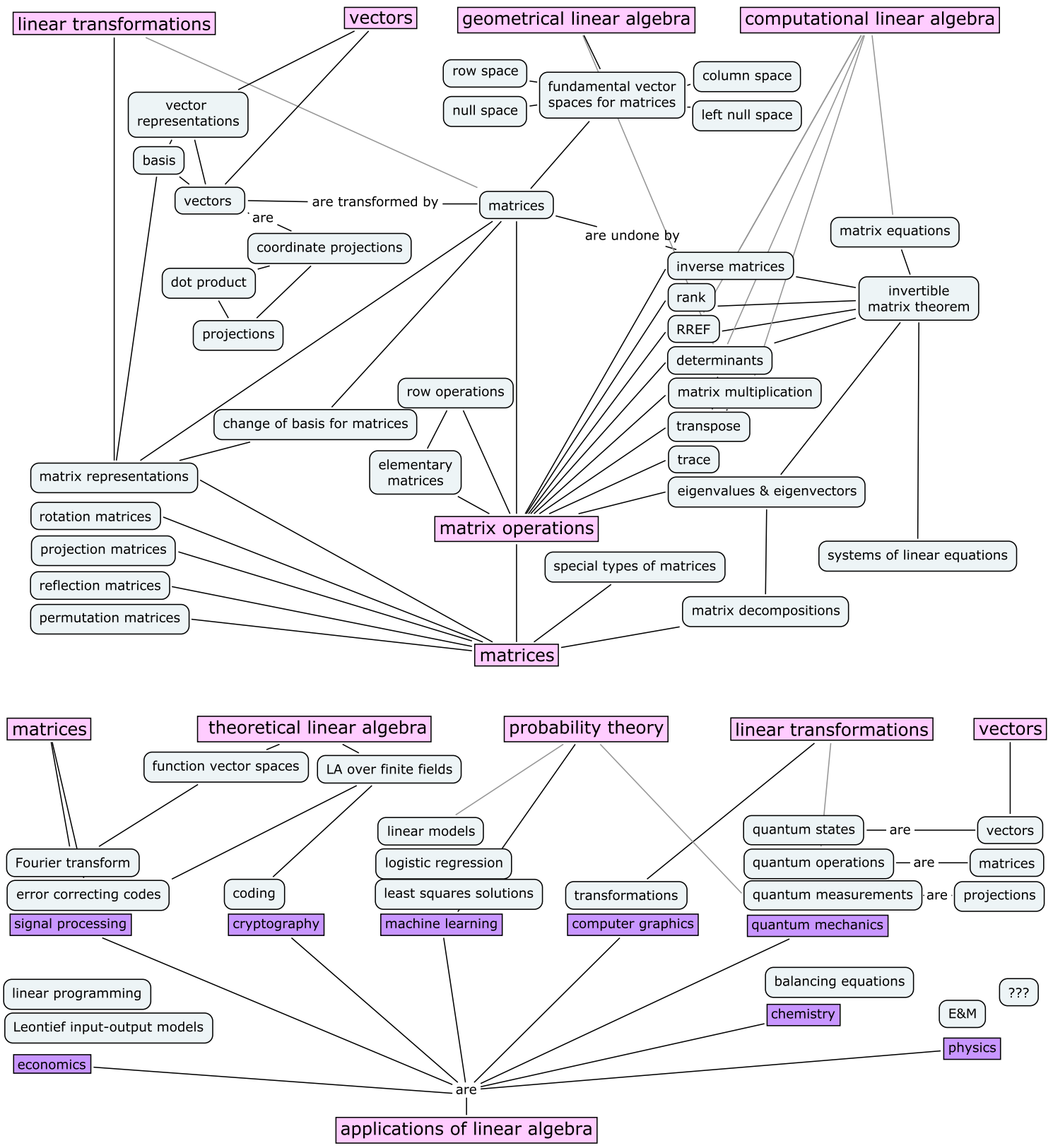

numbers
refer to
variables
are used in
equations
y=mx+b
quadratic eqn
y=ax²+bx+c
x²
√x
|x|
exp(x)
ln(x)
f(x-h)+k
sin(x)
asin(x)
cos(x)
acos(x)
tan(x)
atan(x)
triangle
trig identities
trigonometry
degrees
radians
circle
parallelepiped
ellipse
hyperbola
parallelogram
solve
systems of equations
function inverse
functions
function graphs
are plotted in the
cartesian plane
algebra
high school math
geometry
matrices
row operations
can be repre-
sented as
elementary
matrices
are used in
Gauss-Jordan elimination
to obtain the
RREF
augmented
matrices
systems of
linear quations
can be represented as
adjugate
matrix
matrix equations
determinants
matrix inverse
matrix multiplication
computational linear algebra
coordinate systems
change of basis
=
components
have
expressed w.
respect to a
basis
vectors
are
directions
have a
length
vector operations
projections
used for
dot product
vector products
cross product
vectors
are similar to
complex numbers
geometry
matrices
eigenspaces
row space
column space
null space
left null space
matrix fundamental
vector spaces
vectors components
projections
cooridnate projections
orthogonality
basis
dot product
determinants
areas
2x2 determinant
span
linear independence
volumes
3x3 determinant
vector spaces
abstract inner product spaces
planes
contain
systems of
linear quations
abstract vector spaces
examples:
lines
intersection
of planes
solution
contain
number of
dimensions
examples:
polynomials
matrices
functions
points
geometrical linear algebra
theoretical linear algebra


vectors
functions
matrices
numbers
are transformed by
functions
are undone by
inverse functions
linearity
y=mx
z=mx+ny
preimage
input and output spaces
image
kernel
kernel of adjoint
vectors
are transformed by
linear transformations
matrix representations
are
linear combinations
of the elements of a
basis
are undone by
inverse transformation
rotations
projections
reflections
permutations
vector representation
linear transformations


functions
are transformed by
operators
are undone by
inverse operators
e.g.
integral operator
is undone by
derivative operator
calculus
matrices
input and output spaces
fundamental vector spaces for matrices
eigendecomposition
eigenvalues
is a type of
eigenvectors
points
lines
planes
hyperplanes
matrix decomposition
Gram-Schmidt orthogonalization
vectors
are
vector spaces
can be defined as
span
basis
number of dimensions
linear independence
RREF
can be generalized to
abstract vector spaces
abstract inner product spaces
diagonal
triangular
symmetric
normal
examples:
polynomials
matrices
functions
orthogonal
positive
linear algebra over the complex numbers
special types of matrices
linear algebra over a finite field
theoretical linear algebra


linear transformations
vectors
geometrical linear algebra
computational linear algebra
row space
null space
fundamental vector spaces for matrices
column space
left null space
vector representations
basis
vectors
are transformed by
matrices
are
coordinate projections
are undone by
matrix equations
inverse matrices
dot product
rank
invertible matrix theorem
projections
RREF
determinants
row operations
matrix multiplication
change of basis for matrices
transpose
trace
elementary matrices
matrix representations
eigenvalues & eigenvectors
rotation matrices
matrix operations
projection matrices
systems of linear equations
special types of matrices
reflection matrices
permutation matrices
matrix decompositions
matrices
matrices
theoretical linear algebra
probability theory
linear transformations
vectors
function vector spaces
LA over finite fields
linear models
quantum states
are
vectors
Fourier transform
logistic regression
quantum operations
are
matrices
error correcting codes
coding
least squares solutions
transformations
quantum measurements
are
projections
signal processing
cryptography
machine learning
computer graphics
quantum mechanics
balancing equations
???
linear programming
chemistry
E&M
Leontief input-output models
physics
economics
are
applications of linear algebra